

Documentation of Data Linkage between MIDUS Refresher 2 (MR2) Survey and O*NET 2023 (28.0) Database

This document provides details on the data linkage performed between the MIDUS Refresher 2 (MR2) Survey sample and the 2023 O*NET 28.0 database that produced the following standalone dataset:

MR2_ONET2023_N1247_20260630.sav

Specifically, this dataset was derived by linking MR2 cases' survey responses (collected between 2022-24) and their resulting 2018 Standard Occupation Classification (SOC) codes with the 2023 Occupational Information Network (O*NET 28.0) data release, comprised of 13 distinct files. The Documentation of Constructed Variables for the MIDUS Refresher 2 (MR2) Survey and the O*NET 2023 (28.0) Database Linkage provides a description of each data file.

The MIDUS Refresher 2 sample used SOC codes from 1,247 valid cases to link with the O*NET database.

- If a respondent was not currently working or did not provide adequate occupational information, the case was excluded from the MIDUS-O*NET linkage.
- Military occupation codes were also excluded from the linkage since they are not included in the O*NET database.

The Linkage

SOC codes for MR2 cases were classified by UW Survey Center Staff to six digits in the '12-3456' format. The O*NET uses eight-digit SOC codes in the format '12-3456.78,' including a 2-digit decimal extension. To reconcile this difference, each O*NET SOC code was split into two parts: one containing the six digits before the decimal point and another containing the two digits after it. The 2-digit extension was only used in performing the linkage; it was not included in the SOC codes of the final dataset.

The actual linkage was performed in two steps:

- Step 1 – *for matched cases*: If a MR2 SOC code directly matched the first six digits of an O*NET SOC code, and the O*NET SOC extension code was coded '.00'. The assumption was that the MR2 code '12-3456' is equivalent to the O*NET code '12-3456.00'.
 - 70% (n=873) of the MR2 SOC codes were directly matched to SOC codes contained in the O*NET 28.0 data release.

- Note: The description differs from prior O*NET Linkage data releases, in which we provided a table detailing variation in direct matches across the 13 data files. It is described narratively here because of near unanimity.
 - **Step 2 – for unmatched cases:** There was no direct match between MR2 and O*NET SOC codes for some cases.
 - 30% (n=374) of MR2 SOC codes were unable to match with the SOC codes contained in the O*NET 28.0 data release directly.
 - For these cases, a series of distinct kinds of mean value substitution adjustments was used.
1. **Scenario 1:** If a parent code ‘.00’ was missing from the O*NET dataset, but one or more 2-digit extension codes were available (i.e. ‘.01’, ‘.02’), the mean of the scores with the 2-digit extension O*NET values were calculated and substituted into the variables for the MR2 parent ‘.00’ value.
 - For example, MR2 SOC code ‘11-3031’ did not have a direct match ‘11-3031.00’ in the O*NET data, but O*NET did have ‘11-3021.01’ and ‘11-3031.02’. The mean values of the SOC codes ‘11-3031.01’ and ‘11-3031.02’ were used to substitute the scores for ‘11-3031.00’ and match them with MR2 SOC ‘11-3031’.
 2. **Scenario 2:** If the six-digit MR2 SOC code ended with a ‘0’, it was called a broad occupation code; if the last digit was ‘1’, ‘2’, ‘3’, etc. it was called a detailed occupation code. For example, SOC code ‘11-9030’ refers to ‘education administrators,’ ‘11-9031’ refers to ‘education administrators, preschool and childcare center,’ ‘11-9032’ refers to ‘education administrators, elementary and secondary school.’ ‘11-9030’ is a broad occupation code, while ‘11-9031’ and ‘11-9032’ are detailed occupation codes aligned with specific levels of education. When a broad occupation code was not available in an O*NET dataset, but several detailed occupation codes within that broad code were available, the missing broad occupation codes were replaced with the mean of the detailed occupation codes.
 - For example, MR2 SOC code ‘11-9030’ did not match any O*NET SOC codes. ‘11-9030’ is a broad occupation that includes four detailed occupations (‘11-9031’, ‘11-9032’, ‘11-9033’, ‘11-9039’). Therefore, the mean values of scores for available O*NET SOC codes ‘11-9031’, ‘11-9032’, ‘11-9033’ were used to substitute the scores for MR2 SOC code ‘11-9030’.
 3. **Scenario 3:** If a detailed occupation code was missing from the O*NET dataset, but several other detailed occupation codes within the same broad occupation code were available, then the scores of the missing detailed occupation codes were substituted with the mean values of the other detailed occupation codes within the same broad occupation code.

- For example, MR2 SOC code '31-1014' did not match any O*NET codes. The mean values of scores for O*NET SOC codes '31-1011' and '31-1013' were used to substitute the MR2 SOC code '31-1014'.
4. **Scenario 4:** When a 6-digit MR2 SOC code ended with '99,' it was the last code in a broad category and meant to encompass all cases not listed separately in the broad category. For example, '15-1199' refers to 'Computer occupations, all other,' '19-1099' refers to 'Life scientists, all other.' When this type of MR2 SOC code did match an O*NET SOC code, its scores was populated with the mean scores of all the specific detailed occupation codes.
- For example, MR2 SOC code '21-2099' did not match any O*NET SOC codes. We treated this code as '21-2090' and used the mean values of the scores for SOC codes '21-2011' and '21-2021' to substitute SOC code '21-2099'.

Appendix A details all the specific MR2 SOC codes for which substitutions were used, and the O*NET SOC codes used to compute the mean substitution.

Variable Naming

Once linked, variables from the O*NET 2023 datasets were renamed to conform with MIDUS variable naming conventions. The example below illustrates how original O*NET variable names were retained and incorporated into new variable labels.

Variable renaming example

Original O*NET 2023 Variable Name	O*NET-MIDUS Merged Variable Name	O*NET-MIDUS Merged Variable Label
@1.A.1.a.1	RB1ABIM1A1A1	O*NET 2023 Element ID @1.A.1.a.1: Abilities-Important: Oral Comprehension

In this example, for variable 'RB1ABIM1A1A1',

- 'RB1' represents the MIDUS Refresher sample (R) second wave (B) Project 1 (1);
- 'ABIM' represents the O*Net dimension Abilities-Important;
- '1A1A1' represents the O*Net Element ID @1.A.1.a.1 to which it was linked.

Appendix A
Comprehensive list of unmatched SOC codes and the mean substitution adjustments.

Unmatched SOC codes	SOC Codes Used to Compute Mean Substitution Scores
11-2030	11-2011; 11-2021; 11-2022
11-3013	11-3012
11-9030	11-9031; 11-9032; 11-9033
11-9070	11-9071
11-9179	11-9179.01; 11-9179.02
11-9199	11-9199.01; 11-9199.02; 11-9199.08; 11-9199.09; 11-9199.10; 11-9199.11
13-1030	13-1031; 13-1032
13-1070	13-1071; 13-1074; 13-1075
13-1082	13-1081
13-1199	13-1199.04; 13-1199.05; 13-1199.06
13-2020	13-2023
13-2051	13-2052; 13-2053
13-2070	13-2071; 13-2072
15-1230	15-1231; 15-1232
15-1252	15-1251; 15-1253; 15-1254; 15-1255
15-1255	15-1255.01
15-1299	15-1299.01; 15-1299.02; 15-1299.03; 15-1299.08; 15-1299.09
15-2099	15-2099.01
17-1020	17-1021; 17-1022
17-2070	17-2071; 17-2072
17-2110	17-2111; 17-2112
17-2199	17-2199.03; 17-2199.05; 17-2199.06; 17-2199.07; 17-2199.08; 17-2199.09; 17-2199.10; 17-2199.11
17-3028	17-3021; 17-3022; 17-3023; 17-3024; 17-3025; 17-3026; 17-3027; 17-3029
19-1010	19-1011; 19-1012; 19-1013
19-2030	19-2031; 19-2032
19-2099	19-2099.01
19-3039	19-3032; 19-3033; 19-3034
21-1029	21-1021; 21-1022; 21-1023
21-2099	21-2011; 21-2021
23-1020	23-1021; 23-1022; 23-1023
23-2099	23-2093
25-1069	25-1061; 25-1062; 25-1063; 25-1064; 25-1065; 25-1066; 25-1067
25-1199	25-1192; 25-1193; 25-1194
25-2010	25-2011; 25-2012
25-2020	25-2021; 25-2022; 25-2023
25-2030	25-2031; 25-2032
25-2050	25-2051; 25-2057; 25-2058; 25-2059
25-3031	25-3011; 25-3021; 25-3041
25-3099	25-3011; 25-3021; 25-3041
25-4010	25-4011; 25-4012; 25-4013
25-9040	25-9044

Unmatched SOC codes	SOC Codes Used to Compute Mean Substitution Scores
25-9099	25-9021; 25-9031; 25-9044
27-1010	27-1011; 27-1012; 27-1013; 27-1014
27-2091	27-2011; 27-2012; 27-2021; 27-2022; 27-2023; 27-2031; 27-2032; 27-2041; 27-2042
29-1020	29-1021; 29-1022; 29-1023; 29-1024
29-1214	29-1211; 29-1213; 29-1215; 29-1216; 29-1217; 29-1218
29-1229	29-1229.01; 29-1229.02; 29-1229.03; 29-1229.04; 29-1229.05; 29-1229.06
29-2010	29-2011; 29-2012
29-2042	29-2011; 29-2012; 29-2031; 29-2032; 29-2033; 29-2034; 29-2035; 29-2051; 29-2052; 29-2053; 29-2055; 29-2056; 29-2057; 29-2061; 29-2081; 29-2091; 29-2092; 29-2099
29-2072	29-9091; 29-9092; 29-9093; 29-9099
29-2090	29-2091; 29-2092; 29-2099
29-9021	29-9091; 29-9092; 29-9093; 29-9099
31-9090	31-9091; 31-9092; 31-9093; 31-9094; 31-9095; 31-9096; 31-9097; 31-9099
33-1091	33-1011; 33-1012; 33-1021
33-9030	33-9031; 33-9032
33-9094	33-9091; 33-9092; 33-9093; 33-9099
35-2010	35-2011; 35-2012; 35-2013; 35-2014; 35-2015
35-9099	35-9011; 35-9021; 35-9031
37-2010	37-2011; 37-2012
39-3099	39-3091; 39-3092; 39-3093
39-7010	39-7011; 39-7012
41-2010	41-2011; 41-2012
41-3091	41-3011; 41-3021; 41-3031; 41-3041
41-4010	41-4011; 41-4012
41-9020	41-9021; 41-9022
41-9099	41-9091
43-3099	43-3011; 43-3021; 43-3031; 43-3041; 43-3051; 43-3061; 43-3071
43-4199	43-4111; 43-4121; 43-4131; 43-4141; 43-4151; 43-4161; 43-4171; 43-4181
43-9199	43-9111
45-2090	45-2091; 45-2092; 45-2093
47-2070	47-2071; 47-2072; 47-2073
49-2020	49-2021; 49-2022
49-2090	49-2091; 49-2092; 49-2093; 49-2094; 49-2094; 49-2096; 49-2097; 49-2098
49-3040	49-3041; 49-3042; 49-3043
49-9040	49-9041; 49-9043; 49-9044; 49-9045
49-9060	49-9061; 49-9062; 49-9063; 49-9064
49-9090	49-9091; 49-9092; 49-9094; 49-9095; 49-9096; 49-9097; 49-9098; 49-9099
51-2099	51-2011; 51-2021; 51-2022; 51-2023; 51-2031; 51-2041; 51-2051; 51-2061; 51-2092
51-4120	51-4121; 51-4122
51-4199	51-4191; 51-4192; 51-4193; 51-4194
51-6050	51-6051; 51-6052
51-9010	51-9011; 51-9012
51-9020	51-9123; 51-9124
51-9160	51-9161; 51-9162

Unmatched SOC codes	SOC Codes Used to Compute Mean Substitution Scores
51-9199	51-9191; 51-9192; 51-9193; 51-9194; 51-9195; 51-9196; 51-9197; 51-9198
53-1049	53-1041; 53-1042; 53-1043
53-2010	53-2011; 53-2012
53-2020	53-2021; 53-2022
53-3030	53-3031; 53-3032; 53-3033
53-3051	53-3052
53-3053	53-3052
53-3099	53-3011; 53-3031; 53-3032; 53-3033; 53-3052
53-5020	53-5021; 53-5022
53-7199	53-7121